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API Service

v1.2.2

Base URLs:

- [HTTPS: https://server-wd.ai-ec.cloud:8443](https://server-wd.ai-ec.cloud:8443)

API Usage Instructions:

1.Authorization Acquisition:

Contact our company to obtain your `companyCode` (unique identifier) and `key` (unique secret key). The `companyCode` will be used as an identifier, and the `key` will be used for signing.

Example:

`companyCode: ab56b....f89985d4b786`

`key: 2a1891544dbcf8....b36d03187485e`

2.Signature Generation:

Signature Algorithm:

Use the MD5 algorithm for signature authentication (an example code snippet for the MD5 signature can be found in the following documentation).

Signature String Composition:

First Part:

Sort the parameters (excluding time, which is used in the second part) by their parameter names in Unicode encoding order from smallest to largest. Then concatenate them in the format of `key=value&key=value&`.

Second Part:

Concatenate the time and the assigned key value:

`time: time` is the current timestamp in seconds for UTC+0 timezone.

`key: key` is your unique secret key.

Example:

`time=1723720871&key=2a1891544dbcf8e8b45b36d03187485e`

Complete Example:

Concatenate both parts:

`datalogSn=SXDID888888&storageSn=SXDID888888XXXXXX&time=1723720871&key=2a1891544dbcf8e8b45b36d03187485e`

Finally

use the MD5 signature algorithm (an example code snippet for the MD5 signature can be found in the following documentation) to sign the concatenated string, which will be passed as a parameter when accessing the API.

Server Access:

- 1.First, pass the assigned `companyCode` in the Headers (parameter name is `companyCode`).
- 2.When calling the API, in addition to passing necessary parameters, you also need to pass the time and sign parameters. time is the current timestamp in seconds for UTC+0 timezone (must match the one used in the signature), and sign is the signed data.

Note:

Ensure that the **time** parameter matches the one used in the signature.

Example:

```
// Request Parameters
Map<String, String> param = new HashMap<String, String>();
param.put("dataLogSn", "SXDID888888");
param.put("deviceSn", "SXDID888888XXXXXX");
// Current timestamp in seconds for UTC+0 timezone, must match the one used in
the signature
param.put("time", time);
// signature
param.put("sign", sign);
String result = "";
try {
    result = doPost(HTTP_URL, readParams(param), 3);
} catch (Exception e) {
    result = "";
    e.printStackTrace();
    return new AjaxDto(1);
}
```

3.Supplementary Information:

Sorting According to Unicode Encoding Order:

You can use the default sorting of a `TreeMap`, where parameter names are keys and parameters are values. Iterate through the `TreeMap` to concatenate the string.

Example:

```
// Create a new TreeMap object
TreeMap<String, String> map = new TreeMap<>();
//Store the parameters with the parameter names as keys and parameter values as
values; TreeMap sorts by the Unicode encoding order of the keys by default.
map.put("dataLogSn", "SXDID888888");
map.put("deviceSn", "SXDID888888XXXXXX");
// Get all keys and values
Set<Map.Entry<String, String>> entries = map.entrySet();
// Concatenation example
StringBuffer sign = new StringBuffer();
for (Map.Entry<String, String> entry : entries) {
    //Concatenate
    text.append(entry.getKey() +"="+ entry.getValue());
    text.append("&");
}
```

```

        // Output the map iteratively in ascending order based on the key
        System.out.println("key:" + entry.getKey() + "--value:" +
entry.getValue());
    }
    System.out.println(sign);
    // Output:  datalogSn=SXDID888888&storageSn=SXDID888888XXXXXX&

/**
 * The output shows that `datalogSn` precedes `storageSn` in the iteration.
 * key:datalogSn--value:SXDID888888
 * key:storageSn--value:SXDID888888XXXXXX
 */

```

MD5 Signature:

Algorithm Example:

```

// Package: import org.springframework.util.DigestUtils;

// Input: The string to be signed.
public static String calculateMD5(String input){

    String sign = "";
    try {
        sign =
DigestUtils.md5DigestAsHex(input.getBytes(StandardCharsets.UTF_8));
        // Return the signed result
        return sign;
    }catch (Exception e){
        e.printStackTrace();
    }
    return null;
}

```

Example Return Value: 9ed6aa7f64c574ce7d18f8b47d1bcb6f

API return code

RESULT	DESCRIBE
0	Success
10000	No login or no companyCode
10001	Signature exception
20001	Submitted data cannot be empty
20002	Abnormal permissions
20003	Time format error
20008	Device off-line
20009	Device not exist
60000	SecretKey not exist
60001	Account not enabled

Jave demo

<https://monitor.ai-ec.cloud:8443/resources/upload/GetDataTest.java>

Open Service/Device Service

POST Get basic device information

POST /openApi/device/getBasicsInfo

Body Request parameters

```
{
  "deviceSn": "HS-2408200073-00027",
  "time": "1725450897",
  "sign": "e83ba9c021edd831ae69033f77528ac5"
}
```

Request parameters

Name	Position	Type	Required	Describe	Describe
companyCode	header	string	No		none
Accept-Language	header	string	No		none
body	body	object	No		none
» deviceSn	body	string	Yes	Device Sn	none
» time	body	string	Yes	Second level timestamp (UTC+0)	none
» sign	body	string	Yes	Signature Information	none

Return example

success

```
{
  "result": 0,
  "msg": "Request successfully.",
  "obj": {
    "deviceSn": "HS-2408200073-00027",
    "dataLogSn": "SXDBJG0002",
    "deviceState": 0,
    "deviceVo": {
      "sourceSN": "HS-2408200073-00027",
      "outPriority": 3,
      "outVoltSet": 230,
      "outFreqSet": 50,
      "lineRangeSet": 0,
      "paraModeSet": 0,
      "ecoEn": 0,
      "chgPriority": 2,
      "maxChgCurrSet": 40,
      "chgCurrByPvSet": 16,
      "chgCurrByLineSet": 15,
      "commIdSet": 1,
      "muteEn": 0,
      "onGirdSet": 1,
      "chgFullCurrSet": 3,
      "nG_FuncEn": 0,
      "leakageCheckEn": 0,
      "userPassword": 0,
      "passwordSet": 0,
      "presentTime": "2024-09-04 13:33:28",
      "ctrlFlag1": 144,
      "battActiveSet": 0,
      "onGridActivePowerSet": 800,
      "onGridReactivePowerSet": 0,
      "hybirdPriorityEn": 0,
      "onGridPFSet": 1,
      "gridStandardSet": 4,
      "isoDetectEn": 0,
      "lowIsoValue": 20,
      "extCtGetHostEn": 0,
      "outPhaseDiffSet": 0,

```

```

"powerTurboEn": 0,
"gridConnectVoltMax": 109,
"gridConnectVoltMin": 86,
"gridConnectFreqMax": 50.05,
"gridConnectFreqMin": 47.55,
"gridReConnectVoltMax": 109,
"gridReConnectVoltMin": 86,
"gridReConnectFreqMax": 50.05,
"gridReConnectFreqMin": 47.55,
"gridOVLevel1": 110,
"gridOVDelayTime1": 0.2,
"gridOVLevel2": 125,
"gridOVDelayTime2": 0.16,
"gridUVLevel1": 80,
"gridUVDelayTime1": 3.04,
"gridUVLevel2": 45,
"gridUVDelayTime2": 0.32,
"ovl0minProtectValue": 110,
"gridOFLevel1": 51.5,
"gridOFDelayTime1": 0.12,
"gridOFLevel2": 51.5,
"gridOFDelayTime2": 0.12,
"gridUFLLevel1": 47.5,
"gridUFLDelayTime1": 0.12,
"gridUFLLevel2": 47.5,
"gridUFLDelayTime2": 0.12,
"startOnGridDelay": 1,
"loadingsSlope": 200,
"onGridReconnectDelay": 1,
"reLoadingsSlope": 200,
"powerDynamicRate": 300,
"islandEn": 1,
"hVRTEn": 0,
"lVRTEn": 0,
"constReActPowerEn": 0,
"constPfEn": 0,
"f_PCurveEn": 0,
"p_QCurveEn": 0,
"u_PCurveEn": 0,
"u_QCurveEn": 0,
"dRMEn": 0,
"remoteTurnOffEn": 0,
"permitservice": 1,
"batType": 6,
"cvVolt": 56.8,
"floatVolt": 56.8,
"battOverVolt": 60,
"cvChgTimeSet": 120,
"cvChgBackVolt": 53.6,
"inv2LineVolt": 49.2,
"lineBack2InvVolt": 57.6,
"battLowVolt": 49.6,
"battDelayOffVolt": 48.8,
"battEodVolt": 46.4,
"battDelayOffTime": 30,
"battEodBackVolt": 52.8,
"bmsSet": 0,
"bmsProtocal": 0,

```

```
"batSocLowAlarm": 15,  
"batStopSOC": 5,  
"battFullSOC": 100,  
"inv2LineSOC": 10,  
"lineBack2InvSOC": 100,  
"onTimeChargeEn": 0,  
"chargeStartTime1": "00:00",  
"chargeEndTime1": "00:00",  
"chargeStartTime2": "00:00",  
"chargeEndTime2": "00:00",  
"chargeStartTime3": "00:00",  
"chargeEndTime3": "00:00",  
"onTimeDischgEn": 0,  
"dischgStartTime1": "00:00",  
"dischgEndTime1": "23:00",  
"dischgStartTime2": "00:00",  
"dischgEndTime2": "00:00",  
"dischgStartTime3": "00:00",  
"dischgEndTime3": "00:00",  
"battSocSet": 0,  
"battPackNotUnion": 1,  
"forcechgStopSoc": 5,  
"fP_OF_start": 50.2,  
"fP_UF_star": 49.8,  
"fP_OF_End": 51.5,  
"fP_UF_End": 47.5,  
"fP_OF_DropSlope": 5,  
"fP_UF_LodingSlope": 5,  
"fP_ActPowerRespTr": 1,  
"pQ_P1": 20,  
"pQ_Q1": 0,  
"pQ_P2": 50,  
"pQ_Q2": 0,  
"pQ_P3": 100,  
"pQ_Q3": -31,  
"pQ_P1n": -20,  
"pQ_Q1n": 0,  
"pQ_P2n": -50,  
"pQ_Q2n": 0,  
"pQ_P3n": -100,  
"pQ_Q3n": 44,  
"pQ_EnableVot": 85,  
"pQ_DisableVot": 109,  
"uP_V1": 105,  
"uP_P1": 100,  
"uP_V2": 110,  
"uP_P2": 20,  
"uP_ActPowerRespTr": 10,  
"uQ_Vref": 100,  
"uQ_V1": 92,  
"uQ_Q1": 25,  
"uQ_V2": 96,  
"uQ_Q2": 0,  
"uQ_V3": 105,  
"uQ_Q3": 0,  
"uQ_V4": 108,  
"uQ_Q4": -25,  
"uQ_ReActPowerRespTr": 10,
```

```

    "uQ_VrefAutoEn": 0,
    "uQ_VrefRespTr": 10,
    "hVRT_Level1": 110,
    "hVRT_Level1Time": 70,
    "hVRT_Level2": 116,
    "hVRT_Level2Time": 7,
    "hVRT_Level3": 121,
    "hVRT_Level3Time": 0.14,
    "lVRT_Level1": 85,
    "lVRT_Level1Time": 305,
    "lVRT_Level2": 50,
    "lVRT_Level2Time": 1.8,
    "lVRT_Level3": 15,
    "lVRT_Level3Time": 0.18,
    "modelCode": 0,
    "machCode": 20,
    "productType": 0,
    "manufactureName": "",
    "hwVer_M1": 603,
    "hwVer_M2": 104,
    "hwVer_S1": 0,
    "hwVer_S2": 0,
    "protocolVer": 0,
    "bmsVer": 0,
    "libver": 786432,
    "cpuBuidTime": "Aug 20 2024 11:08:17",
    "customSN": "HS-2408200073-00027",
    "rated_OutputW": 0,
    "maxW_AtInjectVar": 0,
    "maxPF_InjectVar": 0,
    "maxW_AtAbsorbVar": 0,
    "maxPF_AtAbsorbVar": 0,
    "maxOutputVA": 0,
    "powerCtrlLevel": 0,
    "rideThroughLevel": 0,
    "maxInjectVar": 0,
    "maxAbsorbVar": 0,
    "rated_Chgw": 0,
    "rated_ChgVA": 0,
    "ratedGridVlt": 0,
    "maxGridVlt": 0,
    "minGridVlt": 0,
    "gridCtrlModeFlag": 0,
    "reactiveSusceptance": 0,
    "iANA_EnterpriseID": 0,
    "onOffCtrl": 0,
    "restartCtrl": 0,
    "chgCtrl": 0
  }
}
}

```

Return result

Status code	Meaning of Status Code	Describe	Data model
200	OK	success	Inline

Return data structure

Status code **200**

Name	Type	Required	Constraint	Describe	Describe
» result	integer	true	none		none
» msg	string	true	none		none
» obj	object	true	none		none

POST Get real-time device information

POST /openApi/device/getRealTimeInfo

Body Request parameters

```
{
  "deviceSn": "HS-2408200073-00027",
  "time": "1725450897",
  "sign": "e83ba9c021edd831ae69033f77528ac5"
}
```

Request parameters

Name	Position	Type	Required	Describe	Describe
companyCode	header	string	No		none
Accept-Language	header	string	No		none
body	body	object	No		none
» deviceSn	body	string	Yes	Device Sn	none
» time	body	string	Yes	Second level timestamp (UTC+0)	none
» sign	body	string	Yes	Signature Information	none

Return example

success

```
{
  "result": 0,
  "msg": "Request successfully.",
  "obj": {
    "deviceSn": "HS-2408200073-00027",
    "dataLogSn": "SXDBJG0002",
  }
}
```

```

"deviceState": 0,
"devicevo": {
  "lastUpdateTime": null,
  "lastUpdateTimeStamp": 0,
  "updateTime": null,
  "updateTimeStamp": 0,
  "battVolt": 0,
  "battCurr": 0,
  "battSoc": 0,
  "battChgPower": 0,
  "lineChgCurr": 0,
  "pvChgCurr": 0,
  "pv1Volt": 0,
  "pv1Curr": 0,
  "pv1Power": 0,
  "pv2Volt": 0,
  "pv2Curr": 0,
  "pv2Power": 0,
  "outVoltA": 0,
  "outCurrA": 0,
  "outFreq": 0,
  "invCurrA": 0,
  "outW_A": 0,
  "outVA_A": 0,
  "loadPercentA": 0,
  "normalLoadW_A": 0,
  "outVoltB": 0,
  "outCurrB": 0,
  "invCurrB": 0,
  "outW_B": 0,
  "outVA_B": 0,
  "loadPercentB": 0,
  "normalLoadW_B": 0,
  "outVoltC": 0,
  "outCurrC": 0,
  "invCurrC": 0,
  "outW_C": 0,
  "outVA_C": 0,
  "loadPercentC": 0,
  "normalLoadW_C": 0,
  "outSumW": 0,
  "outSumVA": 0,
  "sumLoadPercent": 0,
  "generatedPowerP_A": 0,
  "generatedPowerP_B": 0,
  "generatedPowerP_C": 0,
  "generatedPowerS_A": 0,
  "generatedPowerS_B": 0,
  "generatedPowerS_C": 0,
  "generatedPowerQ_A": 0,
  "generatedPowerQ_B": 0,
  "generatedPowerQ_C": 0,
  "lineVoltA": 0,
  "lineCurrA": 0,
  "lineFreq": 0,
  "linevoltB": 0,
  "lineCurrB": 0,
  "linevoltC": 0,

```

```
"lineCurrC": 0,
"linePowerVA_A": 0,
"linePowerVA_B": 0,
"linePowerVA_C": 0,
"linePowerW_A": 0,
"linePowerW_B": 0,
"linePowerW_C": 0,
"busVoltP": 0,
"busVoltN": 0,
"temper1": 0,
"temper2": 0,
"temper3": 0,
"temper4": 0,
"sIR": 0,
"fanDuty": 0,
"gridPowerW_A": 0,
"gridPowerW_B": 0,
"gridPowerW_C": 0,
"gridCurr_A": 0,
"gridCurr_B": 0,
"gridCurr_C": 0,
"startupStatus": 0,
"outStatus": 0,
"chgStatus": 0,
"paraMode": 0,
"ecoMode": 0,
"battSelfCheck": 0,
"dryContact1": 0,
"dryContact2": 0,
"passwordStatus": 0,
"sysStateFlag1": 0,
"sysStateFlag2": 0,
"sysStateFlag3": 0,
"sysStateFlag4": 0,
"sysStateFlag5": 0,
"sysStateFlag6": 0,
"sysStateFlag7": 0,
"sysStateFlag8": 0,
"sysAlarmFlag1": 0,
"sysAlarmFlag2": 0,
"sysAlarmFlag3": 0,
"sysAlarmFlag4": 0,
"sysAlarmFlag5": 0,
"sysAlarmFlag6": 0,
"sysAlarmFlag7": 0,
"sysAlarmFlag8": 0,
"derOperationState": 0,
"derConnectState": 0,
"derAlarmStatus": 0,
"derInverterState": 0,
"derBattState": 0,
"remoteCtrlStatus": 0,
"bmsReqChgVolt": 0,
"bmsReqEodVolt": 0,
"bmsReqChgCurr": 0,
"bmsReqDisChgCurr": 0,
"battSumPower": 0,
"extPvSumOutW_A": 0,
```

```

    "extPvSumOutW_B": 0,
    "extPvSumOutW_C": 0,
    "backUpSumOutW_A": 0,
    "backUpSumOutW_B": 0,
    "backUpSumOutW_C": 0,
    "backUpSumOutVA_A": 0,
    "backUpSumOutVA_B": 0,
    "backUpSumOutVA_C": 0,
    "normalSumLoadW_A": 0,
    "normalSumLoadW_B": 0,
    "normalSumLoadW_C": 0,
    "generatedSumPowerP_A": 0,
    "generatedSumPowerP_B": 0,
    "generatedSumPowerP_C": 0,
    "generatedSumPowers_A": 0,
    "generatedSumPowers_B": 0,
    "generatedSumPowers_C": 0,
    "lineSumPowerW_A": 0,
    "lineSumPowerW_B": 0,
    "lineSumPowerW_C": 0,
    "generatedACPowerToday": 0,
    "batChgEnergyToday": 0,
    "batDischgEnergyToday": 0,
    "pvGeneratEnergyToday": 0,
    "usedEnergyToday": 0,
    "workDaysTotal": 0,
    "batChgEnergyTotal": 0,
    "batDischgEnergyTotal": 0,
    "pvGeneratEnergyTotal": 0,
    "usedEnergyTotal": 0,
    "lineChgEnergyTday": 0,
    "loadConsumLineTday": 0,
    "invWorkTimeToday": 0,
    "lineworkTimeToday": 0,
    "powerOnTime": null,
    "lineChgEnergyTotal": 0,
    "loadConsumLineTotal": 0,
    "invWorkTimeTotal": 0,
    "lineworkTimeTotal": 0,
    "generatedACPowerTotal": 0,
    "stacTimeBack": null
  }
}
}

```

Return result

Status code	Meaning of Status Code	Describe	Data model
200	OK	success	Inline

Return data structure

Status code **200**

Name	Type	Required	Constraint	Describe	Describe
» result	integer	true	none		none
» msg	string	true	none		none
» obj	object	true	none		none
»» deviceSn	string	true	none		none
»» dataLogSn	string	true	none		none
»» deviceState	null	true	none	0: offline 1:online	none
»» deviceVo	object	true	none		none

POST Get device permission settings information

POST /openApi/device/getSetInfo

Body Request parameters

```
{
  "deviceSn": "HS-2408200073-00027",
  "time": "1725450897",
  "sign": "e83ba9c021edd831ae69033f77528ac5"
}
```

Request parameters

Name	Position	Type	Required	Describe	Describe
companyCode	header	string	No		none
Accept-Language	header	string	No		none
body	body	object	No		none
» deviceSn	body	string	Yes	Device Sn	none
» time	body	string	Yes	Second level timestamp (UTC+0)	none
» sign	body	string	Yes	Signature Information	none

Return example

success

```
{
  "result": 0,
  "msg": "Request successfully.",
  "obj": {
    "deviceSn": "HS-2408200073-00027",
    "dataLogSn": "SXDBJG0002",
  }
}
```

```

"deviceState": 0,
"devicevo": {
  "modelCode": 0,
  "onOffCtrl": 0,
  "ecoEn": 0,
  "maxChgCurrSet": 40,
  "muteEn": 0,
  "onGirdSet": 1,
  "onGridActivePowerSet": 800,
  "outPriority": 3,
  "chgPriority": 2,
  "outVoltSet": 230,
  "outFreqSet": 50,
  "lineRangeSet": 0,
  "paraModeSet": 0,
  "chgCurrByPvSet": 16,
  "chgCurrByLineSet": 15,
  "batType": 6,
  "chgFullCurrSet": 3,
  "battActiveSet": 0,
  "cvVolt": 56.8,
  "floatVolt": 56.8,
  "cvChgTimeSet": 120,
  "cvChgBackVolt": 53.6,
  "inv2LineVolt": 49.2,
  "lineBack2Invvolt": 57.6,
  "battLowVolt": 49.6,
  "battDelayOffvolt": 48.8,
  "battEodVolt": 46.4,
  "battDelayOffTime": 30,
  "battEodBackVolt": 52.8,
  "batSocLowAlarm": 15,
  "batStopSOC": 5,
  "battFullSOC": 100,
  "inv2LinesOC": 10,
  "lineBack2InvSOC": 100,
  "bmsSet": 0,
  "bmsProtocal": 0,
  "gridStandardSet": 4,
  "hybirdPriorityEn": 0,
  "presentTime": "2024-09-04 13:33:28",
  "onTimeChargeEn": 0,
  "chargeStartTime1": "00:00",
  "chargeEndTime1": "00:00",
  "chargeStartTime2": "00:00",
  "chargeEndTime2": "00:00",
  "chargeStartTime3": "00:00",
  "chargeEndTime3": "00:00",
  "onTimeDischgEn": 0,
  "dischgStartTime1": "00:00",
  "dischgEndTime1": "23:00",
  "dischgStartTime2": "00:00",
  "dischgEndTime2": "00:00",
  "dischgStartTime3": "00:00",
  "dischgEndTime3": "00:00"
}
}
}

```

Return result

Status code	Meaning of Status Code	Describe	Data model
200	OK	success	Inline

Return data structure

Status code **200**

Name	Type	Required	Constraint	Describe	Describe
» result	integer	true	none		none
» msg	string	true	none		none
» obj	object	true	none		none

POST Set device parameter information - single address distribution

POST /openApi/device/setParam

Body Request parameters

```
{
  "deviceSn": "HS-2408200073-00027",
  "paramName": "onOffCtrl",
  "paramValue": "1",
  "time": "1725452930",
  "sign": "a6965b6052d82dec3d6131da443a0c4b"
}
```

Request parameters

Name	Position	Type	Required	Describe	Describe
companyCode	header	string	No		none
Accept-Language	header	string	No		none
body	body	object	No		none
» deviceSn	body	string	Yes	Device Sn	none
» paramName	body	string	Yes	Name	none
» paramValue	body	string	Yes	value	none
» time	body	string	Yes	Second level timestamp (UTC+0)	none
» sign	body	string	Yes	Signature Information	none

Return example

success

```
{
  "result": 20008,
  "msg": "Device offline.",
  "obj": null
}
```

```
{
  "result": 1,
  "msg": "setting timeout.",
  "obj": null
}
```

Return result

Status code	Meaning of Status Code	Describe	Data model
200	OK	success	Inline

Return data structure

Status code **200**

Name	Type	Required	Constraint	Describe	Describe
» result	integer	true	none	State	none
» msg	string	true	none	Describe	none
» obj	null	true	none		none

POST Set device parameter information - Multiple address distribution

POST /openApi/device/setCustomParams

Body Request parameters

```
{
  "deviceSn": "HS-2408200073-00027",
  "object": "
{\"chargeEndTime1\": \"12:12\", \"chargeEndTime2\": \"12:12\", \"chargeEndTime3\": \"12:12\", \"chargeStartTime1\": \"12:12\", \"chargeStartTime2\": \"12:12\", \"chargeStartTime3\": \"12:12\", \"dischgEndTime1\": \"12:12\", \"dischgEndTime2\": \"12:12\", \"dischgEndTime3\": \"12:12\", \"dischgStartTime1\": \"12:12\", \"dischgStartTime2\": \"12:12\", \"dischgStartTime3\": \"12:12\", \"onTimeChargeEn\": 1, \"onTimeDischgEn\": 1}",
  "sign": "134bc99ab7bcf774c34e658ca7f45d5a",
  "time": "1725454716"
}
```

Request parameters

Name	Position	Type	Required	Describe	Describe
companyCode	header	string	No		none
Accept-Language	header	string	No		none
body	body	object	No		none
» deviceSn	body	string	Yes	Device Sn	none
» object	body	string	Yes	JSON string of set parameters	none
» time	body	string	Yes	Second level timestamp (UTC+0)	none
» sign	body	string	Yes	Signature Information	none

Return example

success

```
{
  "result": 20008,
  "msg": "Device offline.",
  "obj": null
}
```

Return result

Status code	Meaning of Status Code	Describe	Data model
200	OK	success	Inline

Return data structure

Status code **200**

Name	Type	Required	Constraint	Describe	Describe
» result	integer	true	none	State	none
» msg	string	true	none	Describe	none
» obj	null	true	none		none

Data model

Rated data model

Field name	Required	Type	Describe	Unit
sourceSN		String	Product Serial Number	
outPriority		Integer	Load Supply Priority	
outVoltSet		Float	System Voltage Setting	V
outFreqSet		Float	System Frequency Level	Hz
lineRangeSet		Integer	Utility range setting	
paraModeSet		Integer	Parallel mode setting	
ecoEn		Integer	Energy saving mode enable	
chgPriority		Integer	Charging priority setting	

Field name	Required	Type	Describe	Unit
maxChgCurrSet		Float	Maximum charging current for stand-alone machine	A
chgCurrByPvSet		Float	PV maximum charging current	A
chgCurrByLineSet		Float	Maximum charging current of utility power	A
commIdSet		Integer	Local communication address setting	
muteEn		Integer	Buzzer mute enable	
onGridSet		Integer	Energy internet setting	
chgFullCurrSet		Float	Battery full judgment current setting	A
nG_FuncEn		Integer	Inverter mode N-G connection enable	
leakageCheckEn		Integer	Leakage current detection enable	
userPassWord		Integer	User password setting value	
passWordSet		Integer	User input password value	
presentTime		String	Machine current time	
ctrlFlag1		Integer	Control flag bit 1	
battActiveSet		Integer	Battery activation enable	
onGridActivePowerSet		Integer	Grid active power setting	W
onGridReactivePowerSet		Integer	Grid-connected reactive power setting	Var
hybirdPriorityEn		Integer	Grid mixing priority enable	
onGridPFSet		Float	Grid power factor setting	
gridStandardSet		Integer	Grid standard setting	
iSODetectEn		Integer	Insulation impedance detection enable	
lowIsoValue		Integer	Insulation impedance low threshold	kΩ
extCtGetHostEn		Integer	External CT reading host	
outPhaseDiffSet		Integer	Output phase setting	
powerTurboEn		Integer	Power-Turbo enable	

Field name	Required	Type	Describe	Unit
gridConnectVoltMax		Integer	Grid connection voltage upper limit	V
gridConnectVoltMin		Integer	Grid connection voltage lower limit	V
gridConnectFreqMax		Float	Grid connection frequency upper limit	Hz
gridConnectFreqMin		Float	Grid connection frequency lower limit	Hz
gridReConnectVoltMax		Integer	Grid reconnect voltage upper limit	V
gridReConnectVoltMin		Integer	Grid reconnection voltage lower limit	V
gridReConnectFreqMax		Float	Upper limit of grid reconnection frequency	Hz
gridReConnectFreqMin		Float	Lower limit of reconnection frequency	Hz
gridOVLevel1		Integer	First-order overvoltage trigger value	V
gridOVDelayTime1		Float	First-order overvoltage delay time	S
gridOVLevel2		Integer	Second-order overvoltage trigger value	V
gridOVDelayTime2		Float	Second-order overvoltage delay time	S
gridUVLevel1		Integer	First-order undervoltage trigger value	V
gridUVDelayTime1		Float	First order undervoltage delay time	S
gridUVLevel2		Integer	Second-order undervoltage trigger value	V
gridUVDelayTime2		Float	Second-order undervoltage delay time	S
oV10minProtectValue		Integer	10min overvoltage protection value	V
gridOFLevel1		Float	First-order overfrequency trigger value	Hz
gridOFDelayTime1		Float	First-order overfrequency delay time	S

Field name	Required	Type	Describe	Unit
gridOFLevel2		Float	Second-order overfrequency trigger value	Hz
gridOFDelayTime2		Float	Second-order overfrequency delay time	S
gridUFLLevel1		Float	First-order underfrequency trigger value	Hz
gridUFDelayTime1		Float	First order underfrequency delay time	S
gridUFLLevel2		Float	Second order underfrequency trigger value	Hz
gridUFDelayTime2		Float	Second-order underfrequency delay time	S
startOnGridDelay		Float	Grid-connected start-up waiting time	S
loadingSlope		Integer	Grid power rise slope	%
onGridReconnectDelay		Float	Grid fault reconnection waiting time	S
reLoadingSlope		Integer	Grid reconnection power rising slope	%
powerDynamicRate		Integer	Grid power dynamic slope	%
islandEn		Integer	Anti-islanding enable	
hVRTEn		Integer	HV Ride-through Enable	
IVRTEn		Integer	Low voltage ride-through enable	
constReActPowerEn		Integer	Fixed reactive power grid connection enable	
constPfEn		Integer	Fixed PF grid connection enable	
f_PCurveEn		Integer	F-P curve enable	
p_QCurveEn		Integer	P-Q curve enable	
u_PCurveEn		Integer	U-P curve enable	
u_QCurveEn		Integer	U-Q curve enable	
dRMEn		Integer	DRM enable	
remoteTurnOffEn		Integer	Remote shutdown enable	

Field name	Required	Type	Describe	Unit
permitservice		Integer	Inverter service enable	
batType		Integer	Battery type setting	
cvVolt		Float	Constant voltage setting	V
floatVolt		Float	Float charge voltage setting	V
battOverVolt		Float	Battery overvoltage warning voltage setting	V
cvChgTimeSet		Integer	Constant voltage charging time setting	min
cvChgBackVolt		Float	Voltage setting for return to constant voltage charging	V
inv2LineVolt		Float	Battery switching to mains voltage point setting	V
lineBack2InvVolt		Float	Utility to battery voltage point setting	V
battLowVolt		Float	Battery undervoltage alarm point setting	V
battDelayOffVolt		Float	Battery low voltage delay shutdown point setting	V
battEodVolt		Float	Battery EOD voltage setting	V
battDelayOffTime		Integer	Battery voltage delay shutdown time setting	S
battEodBackVolt		Float	Battery EOD clear voltage	V
bmsSet		Integer	BMS Function Setting	
bmsProtocal		Integer	BMS protocol setting	
batSocLowAlarm		Integer	Battery SOC low alarm value setting	
batStopSOC		Integer	Battery SOC low shutdown point setting	
battFullSOC		Integer	Battery full SOC judgment value setting	
inv2LineSOC		Integer	Inverter to mains SOC value setting	
lineBack2InvSOC		Integer	SOC value setting for utility power cut back to inverter.	
onTimeChargeEn		Integer	Timed utility charge enable	

Field name	Required	Type	Describe	Unit
chargeStartTime1		String	1 section start charging time	
chargeEndTime1		String	1-segment end charging time	
chargeStartTime2		String	2-segment start charging time	
chargeEndTime2		String	2-segment end charging time	
chargeStartTime3		String	3-segment start charging time	
chargeEndTime3		String	3-segment end charge time	
onTimeDischgEn		Integer	Timed battery discharge enable	
dischgStartTime1		String	1 segment start time	
dischgEndTime1		String	1-segment end discharge time	
dischgStartTime2		String	2-segment start time	
dischgEndTime2		String	2-segment end discharge time	
dischgStartTime3		String	3-segment start time	
dischgEndTime3		String	3-segment end discharge time	
battSocSet		Integer	Battery SOC setting	%
battPackNotUnion		Integer	Parallel battery pack independent enable	%
forceChgStopSoc		Integer	Battery strong charging stop SOC setting	
fP_OF_Start		Float	FP overfrequency starting point	Hz
fP_UF_Star		Float	FP underfrequency starting point	Hz
fP_OF_End		Float	FP overclocking end point	Hz
fP_UF_End		Float	FP underfrequency end point	Hz
fP_OF_DropSlope		Integer	FP overclocking load down slope	%

Field name	Required	Type	Describe	Unit
fP_UF_LodingSlope		Integer	FP underfrequency loading slope	%
fP_ActPowerRespTr		Float	FP active open loop response time	S
pQ_P1		Integer	pQ_P1 active power	%
pQ_Q1		Integer	PQ_Q1 point reactive power	%
pQ_P2		Integer	PQ_P2 active power	%
pQ_Q2		Integer	PQ_Q2 reactive power	%
pQ_P3		Integer	PQ_P3 active power	%
pQ_Q3		Integer	PQ_Q3 reactive power	%
pQ_P1n		Integer	PQ_P1n point active power	%
pQ_Q1n		Integer	PQ_Q1n point reactive power	%
pQ_P2n		Integer	PQ_P2n point active power	%
pQ_Q2n		Integer	PQ_Q2n point reactive power	%
pQ_P3n		Integer	PQ_P3n point active power	%
pQ_Q3n		Integer	Reactive power at point PQ_Q3n	%
pQ_EnableVolt		Integer	Entering PQ curve voltage	V
pQ_DisableVolt		Integer	Exit PQ curve voltage	V
uP_V1		Integer	Voltage at point UP_V1	%
uP_P1		Integer	UP_P1 active power	%
uP_V2		Integer	Voltage at point UP_V2	%
uP_P2		Integer	UP_P2 active power	%
uP_ActPowerRespTr		Float	UP active open loop response time	S
uQ_Vref		Integer	UQ_Vref voltage	V
uQ_V1		Integer	UQ_V1 voltage	V
uQ_Q1		Integer	UQ_Q1 reactive power	%
uQ_V2		Integer	UQ_V2 voltage	V
uQ_Q2		Integer	UQ_Q2 reactive power	%
uQ_V3		Integer	UQ_V3 voltage	V

Field name	Required	Type	Describe	Unit
uQ_Q3		Integer	UQ_Q3 reactive power	%
uQ_V4		Integer	UQ_V4 voltage	V
uQ_Q4		Integer	UQ_Q4 reactive power	%
uQ_ReActPowerRespTr		Float	UQ reactive open loop response time	S
uQ_VrefAutoEn		Integer	Verf auto-tuning enable	
uQ_VrefRespTr		Float	Vref open loop response time	S
hVRT_Level1		Integer	HV first-order crossing voltage point	V
hVRT_Level1Time		Float	HV first-order crossing delay time	S
hVRT_Level2		Integer	HV second-order crossing voltage point	V
hVRT_Level2Time		Float	HV second-order ride-through delay time	S
hVRT_Level3		Integer	HV third order ride-through voltage point	V
hVRT_Level3Time		Float	HV third order crossing delay time	S
IVRT_Level1		Integer	Low-voltage first-order crossing voltage point	V
IVRT_Level1Time		Float	Low-voltage first-order ride-through delay time	S
IVRT_Level2		Integer	Low-voltage second-order crossing voltage point	V
IVRT_Level2Time		Float	Low-voltage second-order ride-through delay time	S
IVRT_Level3		Integer	Low-voltage third-order crossing voltage point	V
IVRT_Level3Time		Float	Low-voltage third-order crossing delay time	S
modelCode		Integer	1-National standard 2-European standard 3-American standard	
machCode		Integer	Model Code	

Field name	Required	Type	Describe	Unit
productType		Integer	Product Type	
manufactureName		String	Manufacturer Name	
hwVer_M1		Integer	Main Chip Software Version 1	
hwVer_M2		Integer	Master Chip Software Version 2	
hwVer_S1		Integer	Slave Software Version 1	
hwVer_S2		Integer	Slave Software Version 2	
protocolVer		Integer	Modbus Protocol Version	
bmsVer		Integer	BMS version	
libVer		Integer	Software libraries version	
cpuBuidTime		String	Software Compilation Time	
customSN		String	Customer-defined serial number	
rated_OutputW		Integer	Rated active power	W
maxW_AtInjectVar		Integer	Active power at maximum injected reactive power	W
maxPF_InjectVar		Float	Power factor at maximum injected reactive power	
maxW_AtAbsorbVar		Integer	Maximum active power at reactive power absorption	W
maxPF_AtAbsorbVar		Float	Power factor at maximum reactive power absorption	
maxOutputVA		Integer	Maximum apparent discharge power	VA
powerCtrlLevel		Integer	Reactive power and voltage power control capability	
rideThroughLevel		Integer	Voltage and frequency ride-through capability class	
maxInjectVar		Integer	Maximum injected reactive power	Var
maxAbsorbVar		Integer	Maximum amount of absorbed reactive power	Var
rated_ChgW		Integer	Rated charging power	W

Field name	Required	Type	Describe	Unit
rated_ChgVA		Integer	Maximum apparent charging power	W
ratedGridVolt		Integer	Rated AC voltage	V
maxGridVolt		Integer	Maximum permissible AC voltage	V
minGridVlot		Integer	Minimum permissible AC voltage	V
gridCtrlModeFlag		Integer	Supported control mode symbols	
reactiveSusceptance		Integer	Reactive Power Conductivity	S
iANA_EnterpriseID		Integer	IANA manufacturer's number	
onOffCtrl		Integer	Switching machine control	
restartCtrl		Integer	Machine restart control	
chgCtrl		Integer	Off Charge Control	

Real time data model

Field name	Required	Type	Describe	Unit
battVolt		Float	Battery Voltage	V
battCurr		Float	Battery Current	A
battSoc		Integer	Battery remaining power SOC	
battChgPower		Integer	Battery charging power	W
lineChgCurr		Float	Utility Charging Current	A
pvChgCurr		Float	PV Charging Current	A
pv1Volt		Float	1st PV panel voltage	V
pv1Curr		Float	1st PV panel current	A
pv1Power		Integer	1st PV panel output power	W
pv2Volt		Float	2nd PV panel voltage	V
pv2Curr		Float	2st PV panel current	A
pv2Power		Integer	2st PV panel output power	W
outVoltA		Float	A-phase output voltage	V
outCurrA		Float	A-phase output current	A
outFreq		Float	Output frequency	Hz

Field name	Required	Type	Describe	Unit
invCurrA		Float	A-phase inductor current	A
outW_A		Integer	A-phase output active power	W
outVA_A		Integer	A-phase output apparent power	VA
loadPercentA		Integer	A-phase output load ratio	%
normalLoadW_A		Integer	A-phase conventional load active power	W
outVoltB		Float	B-phase output voltage	V
outCurrB		Float	B-phase output current	A
invCurrB		Float	B-phase inductor current	A
outW_B		Integer	B-phase output active power	W
outVA_B		Integer	B-phase output apparent power	VA
loadPercentB		Integer	B-phase output load factor	%
normalLoadW_B		Integer	B-phase conventional load active power	W
outVoltC		Float	C-phase output voltage	V
outCurrC		Float	C-phase output current	A
invCurrC		Float	C-phase inverter inductor current	A
outW_C		Integer	C-phase output active power	W
outVA_C		Integer	C-phase output apparent power	VA
loadPercentC		Integer	C-phase output load factor	%
normalLoadW_C		Integer	C-phase conventional load active power	W
outSumW		Integer	Whole machine output active power	W
outSumVA		Integer	Apparent power output of the whole machine	VA
sumLoadPercent		Integer	Output load ratio of the whole machine	%

Field name	Required	Type	Describe	Unit
generatedPowerP_A		Integer	A-phase inverter active power	W
generatedPowerP_B		Integer	B-phase inverter active power	W
generatedPowerP_C		Integer	Phase C inverter active power	W
generatedPowerS_A		Integer	A-phase inverter apparent power	VA
generatedPowerS_B		Integer	B-phase inverter apparent power	VA
generatedPowerS_C		Integer	C-phase inverter apparent power	VA
generatedPowerQ_A		Integer	Inverter phase A reactive power	Var
generatedPowerQ_B		Integer	Inverter phase B reactive power	Var
generatedPowerQ_C		Integer	Inverter phase C reactive power	Var
lineVoltA		Float	Utility phase A voltage	V
lineCurrA		Float	Utility phase A current	A
lineFreq		Float	Utility frequency	Hz
lineVoltB		Float	Utility phase B voltage	V
lineCurrB		Float	Utility phase B current	A
lineVoltC		Float	Utility phase C voltage	V
lineCurrC		Float	Utility phase C current	A
linePowerVA_A		Integer	Utility phase A apparent power	VA
linePowerVA_B		Integer	utility phase B apparent power	VA
linePowerVA_C		Integer	utility phase C apparent power	VA
linePowerW_A		Integer	Utility phase A active power	W
linePowerW_B		Integer	Utility phase B active power	W
linePowerW_C		Integer	Utility phase C active power	W
busVoltP		Float	Positive bus voltage	V

Field name	Required	Type	Describe	Unit
busVoltN		Float	Negative bus voltage	V
temper1		Float	Temperature sampling point 1 temperature	°C
temper2		Float	Temperature sampling point 2	°C
temper3		Float	Temperature sampling point 3	°C
temper4		Float	Temperature sampling point 4	°C
sIR		Integer	Insulation impedance	KΩ
fanDuty		Integer	Fan speed duty cycle	%
gridPowerW_A		Integer	A-phase grid-side active power	W
gridPowerW_B		Integer	B-phase grid-side active power	W
gridPowerW_C		Integer	Phase C grid-side active power	W
gridCurr_A		Float	A-phase grid-side current	A
gridCurr_B		Float	B-phase grid-side current	A
gridCurr_C		Float	C-phase grid-side current	A
startupStatus		Integer	Machine power-on status	
outStatus		Integer	Output status	
chgStatus		Integer	Charging status	
paraMode		Integer	Machine Parallel Mode	
ecoMode		Integer	Energy saving mode	
battSelfCheck		Integer	Battery self-test status	
dryContact1		Integer	Output dry contact 1 status	
dryContact2		Integer	Output dry contact 2 status	
passwordStatus		Integer	Password protection status symbol	
sysStateFlag1		Integer	Machine Status Bit 1	
sysStateFlag2		Integer	Machine Status Bit 2	
sysStateFlag3		Integer	Machine Status Bit 3	

Field name	Required	Type	Describe	Unit
sysStateFlag4		Integer	Machine Status Bit 4	
sysStateFlag5		Integer	Machine Status Bit 5	
sysStateFlag6		Integer	Machine Status Bit 6	
sysStateFlag7		Integer	Machine Status Bit 7	
sysStateFlag8		Integer	Machine Status Bit 8	
sysAlarmFlag1		Integer	Machine Alarm Bit 1	
sysAlarmFlag2		Integer	Machine Alarm Bit 2	
sysAlarmFlag3		Integer	Machine Alarm Bit 3	
sysAlarmFlag4		Integer	Machine Alarm Bit 4	
sysAlarmFlag5		Integer	Machine Alarm Bit 5	
sysAlarmFlag6		Integer	Machine Alarm Bit 6	
sysAlarmFlag7		Integer	Machine Alarm Bit 7	
sysAlarmFlag8		Integer	Machine Alarm Bit 8	
dEROperationState		Integer	DER operation status	
dERConnectState		Integer	DER grid-connected status	
dERAlarmStatus		Integer	DER Alarm Message	
dERInverterState		Integer	Inverter Status	
dERBattState		Integer	Battery status	
remoteCtrlStatus		Integer	Local/remote control status	
bmsReqChgVolt		Float	BMS request charging voltage	V
bmsReqEodVolt		Float	BMS request discharge limit voltage	V
bmsReqChgCurr		Float	BMS request charging current	A
bmsReqDisChgCurr		Float	BMS request discharge limit current	A
battSumPower		Float	Total battery power	kW
extPvSumOutW_A		Float	Total power generated by external photovoltaic panel phase A	kW
extPvSumOutW_B		Float	Total power of external PV panel phase B	kW

Field name	Required	Type	Describe	Unit
extPvSumOutW_C		Float	Total power of external PV panel phase C	kW
backUpSumOutW_A		Float	Total active power of backup load phase A	kW
backUpSumOutW_B		Float	Total active power of backup load phase B	kW
backUpSumOutW_C		Float	Total active power of backup load phase C	kW
backUpSumOutVA_A		Float	Total apparent power of backup load phase A	KVA
backUpSumOutVA_B		Float	Total apparent power of backup load phase B	KVA
backUpSumOutVA_C		Float	Total apparent power of backup load phase C	KVA
normalSumLoadW_A		Float	Total active power of phase A of conventional load	kW
normalSumLoadW_B		Float	Total active power of phase B of conventional load	kW
normalSumLoadW_C		Float	Total active power of phase C of conventional load	kW
generatedSumPowerP_A		Float	Total active power of inverter phase A	kW
generatedSumPowerP_B		Float	Total active power of inverter phase B	kW
generatedSumPowerP_C		Float	Total active power of inverter phase C	kW
generatedSumPowerS_A		Float	Inverter phase A total apparent power	KVA
generatedSumPowerS_B		Float	Inverter phase B total apparent power	KVA
generatedSumPowerS_C		Float	Inverter phase C total apparent power	KVA
lineSumPowerW_A		Float	Total active power of utility phase A	kW
lineSumPowerW_B		Float	Total active power of utility phase B	kW
lineSumPowerW_C		Float	Total active power of utility phase C	kW

Field name	Required	Type	Describe	Unit
generatedACPowerToday		Float	AC power generation for the day	kWh
batChgEnergyToday		Float	Battery charging power on the same day	kWh
batDischgEnergyToday		Float	Battery discharging power on the same day	kWh
pvGeneratEnergyToday		Float	PV power generation on the same day	kWh
usedEnergyToday		Float	Load power consumption on the same day	kWh
workDaysTotal		Integer	Total operating days	
batChgEnergyTotal		Float	Accumulated Battery Charging	kWh
batDischgEnergyTotal		Float	Accumulated Battery Discharge	kWh
pvGeneratEnergyTotal		Float	Accumulated PV Power Generation	kWh
usedEnergyTotal		Float	Accumulated load power consumption	kWh
lineChgEnergyTday		Float	Charging power from utility on the same day	kWh
loadConsumLineTday		Float	Power consumed by the load from the utility on the same day	kWh
invWorkTimeToday		Integer	Operating time of inverter on the same day	min
lineWorkTimeToday		Integer	Operating time of bypass on the same day	min
powerOnTime		String	Start-up time	
lineChgEnergyTotal		Float	Accumulated charging capacity of utility	kWh
loadConsumLineTotal		Float	Accumulated load consumption from utility	kWh
invWorkTimeTotal		Integer	Cumulative operating time of inverter	h
lineWorkTimeTotal		Integer	Accumulated operating time of bypass	h

Field name	Required	Type	Describe	Unit
generatedACPowerTotal		Float	Accumulated AC Power Generation	kWh
stacTimeBack		String	Statistical data time backup	h